

Database Technology — Reference Paper

Question 1. Database Modeling (40 points)

1.1

Please give an ODL design for the International Diplomatic Relations database which includes the following information:

1. Continent: including its name, population and size.
2. Country: including its name, population, size, GDP, the continent it belongs to, and the countries that have established diplomatic relations with it.
3. RegionalConflict: including its name and the countries involved in it.

Use particular constraints when necessary, and convert it to relational database schemas.

Question 2. Normalization (20 points)

2.1

Consider the relation R with the schema $R(A, B, C, D, E)$, in which

$$B \twoheadrightarrow E, \quad AB \twoheadrightarrow C, \quad BC \rightarrow DE, \quad AB \rightarrow CD$$

hold. Is R in 4NF? If not, decompose it into 4NF beginning with the first violated functional dependency.

2.2

If the former relation is decomposed into $S(A, B, C, E)$ and some other relations, please compute all the completely nontrivial functional dependencies that hold in S and express them in the form of minimal basis.

Question 3. Relational Algebra (10 points)

There is a database used to describe the 13th National People's Congress. The database schemas are:

AnnualSession(ASname : varchar(28), startdate : datetime, enddate : datetime, agenda : varchar(3000))

Deputy(ID : varchar(18), name : varchar(50), gender : char(1), age : integer, ASname : varchar(28))

Bill(number : varchar(10), title : varchar(80), content : text, ASname : varchar(28), ID : varchar(18))

Write one single expression of relational algebra to answer each of the following queries respectively.

3.1

Find out the names of the deputies who are older than 40.

3.2

Find out the names of the deputies who have proposed at least one bill with the title containing "education fairness".

Question 4. SQL Queries (30 points)

There is a database used to describe the 13th National People's Congress. The database schemas are:

AnnualSession(ASname : varchar(28), startdate : datetime, enddate : datetime, agenda : varchar(3000))

Deputy(ID : varchar(18), name : varchar(50), gender : char(1), age : integer, ASname : varchar(28))

Bill(number : varchar(10), title : varchar(80), content : text, ASname : varchar(28), ID : varchar(18))

Write one single SQL statement to answer each of the following questions respectively.

4.1

Find out the total number and the average age of the deputies who are older than 40.

4.2

Find out the total number of the bills for each annual session respectively. Please show the result in the descending order of the total number followed by *ASname*.

4.3

Find out the total number of the bills for each deputy whose age is less than the average age of all deputies. Please show the information including *ASname*, *startdate*, *ID*, *name*, and the number of bills.

Reference Paper Answer

Question 1. Database Modeling (40 points)

1.1

ODL design:

```
interface Continent (key name) {
    attribute string name;
    attribute float population;
    attribute float size;

    relationship Set<Country> members
        inverse Country::belongedTo;
}

interface Country (key name) {
    attribute string name;
    attribute float population;
    attribute float size;
    attribute float GDP;

    relationship Continent belongedTo
        inverse Continent::members;
    relationship Set<Country> leader
        inverse Country::follower;
    relationship Set<Country> follower
        inverse Country::leader;
    relationship Set<RegionalConflict> involvedIn
        inverse RegionalConflict::involve;
}

interface RegionalConflict (key name) {
    attribute string name;

    relationship Set<Country> involve
        inverse Country::involvedIn;
}
```

Relational database schemas converted from the ODL design:

Continent(name, population, size)
Country(name, population, size, GDP, *ContinentName*, *CountryName*)
RegionalConflict(name, *CountryName*)

The E/R-style relational schemas shown in the reference answer are:

Continent(name, population, size)
Country(name, population, size, GDP)
RegionalConflict(name)
belong(*CountryName*, *ContinentName*)
involve(*CountryName*, *RegionalConflictName*)
diplomatic(*CountryName1*, *CountryName2*)

Question 2. Normalization (20 points)

2.1

1. This relation has one key $\{A, B\}$.
2. R is not in 4NF, because there exist nontrivial multivalued dependencies whose left side does not contain a key.
3. Use the 4NF violation $BC \twoheadrightarrow DE$ to decompose R into $R_1(B, C, D, E)$ and $R_2(A, B, C)$.
4. The key of R_1 is $\{B, C\}$, so R_1 is not in 4NF.
5. Use the 4NF violation $B \twoheadrightarrow E$ to decompose R_1 into $R_3(B, E)$ and $R_4(B, C, D)$.
6. R_3 and R_4 are both in 4NF.
7. The key of R_2 is $\{A, B\}$, so R_2 is in 4NF.
8. Therefore, the final decomposition result is $R_3(B, E)$, $R_4(B, C, D)$, and $R_2(A, B, C)$.

2.2

For $S(A, B, C, E)$, compute closures of proper subsets:

$$\begin{aligned}\{A\}^+ &= \{A\}, & \{B\}^+ &= \{B\}, & \{C\}^+ &= \{C\}, & \{E\}^+ &= \{E\} \\ \{A, B\}^+ &= \{A, B, C, D, E\}, & \{A, C\}^+ &= \{A, C\}, & \{A, E\}^+ &= \{A, E\} \\ \{B, C\}^+ &= \{B, C, D, E\}, & \{B, E\}^+ &= \{B, E\}, & \{C, E\}^+ &= \{C, E\} \\ \{A, B, C\}^+ &= \{A, B, C, D, E\}, & \{A, B, E\}^+ &= \{A, B, C, D, E\} \\ \{A, C, E\}^+ &= \{A, C, E\}, & \{B, C, E\}^+ &= \{B, C, D, E\}\end{aligned}$$

The nontrivial functional dependencies holding in S are:

$$AB \rightarrow C, \quad AB \rightarrow E, \quad BC \rightarrow E, \quad ABC \rightarrow E, \quad ABE \rightarrow C$$

The minimal basis is:

$$\{AB \rightarrow C, BC \rightarrow E\}$$

Question 3. Relational Algebra (10 points)

3.1

$$\pi_{name}(\sigma_{age > 40}(Deputy))$$

3.2

$$\pi_{name}(Deputy \bowtie (\sigma_{title \text{ like } \%education\ fairness\ \%}(Bill)))$$

Question 4. SQL Queries (30 points)

4.1

```
SELECT Count(*), Avg(age)
FROM Deputy
WHERE age > 40;
```

4.2

```
SELECT Count(*) AS cnt, ASname
FROM Bill
GROUP BY ASname
ORDER BY cnt DESC, ASname DESC;
```

4.3

```
SELECT Max(a.ASName), Max(a.startdate), b.ID, Max(b.name), Count(*)
FROM AnnualSession a, Deputy b, Bill c
WHERE b.age < (SELECT Avg(age) FROM Deputy)
  AND a.ASName = c.ASName
  AND b.ID = c.ID
GROUP BY b.ID;
```

* The reference answers were transcribed by AI from courseware photos and are for reference only.